23	**A**	$I=\dfrac{E}{R}=\dfrac{E}{\dfrac{\rho L}{A}}=\dfrac{EA}{\rho L}=\dfrac{E\pi\left(d/2\right)^2}{\rho L}\dfrac{\pi d^2E}{4\rho L}$ Hence $I\propto\dfrac{d^2}{\rho}$ since L and E across the wires in parallel are constants. $\dfrac{I_X}{I_Y}=\dfrac{{d_X}^2}{\rho_X}\times\dfrac{\rho_Y}{{d_Y}^2}=\dfrac{\left(\dfrac{1}{4}d_Y\right)^2\times\rho_Y}{{d_Y}^2\times\dfrac{1}{2}\rho_Y}=\dfrac{2}{16}=\dfrac{1}{8}$ $\dfrac{I_X}{I_{total}}=\dfrac{1}{9}$

24	**D**	Since ammeter reading is zero, there is also no current in the middle wire joining the